# Jasmine Gao

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## Education

University of California San Diego, Data Science, Sophomore

- GPA: 3.650

William P. Clements High School, Grade 9-12

- GPA: 101.72 (100 PT weighted)/4.57(weighted)/4.0(unweighted)

## Research & Extracurricular Activities

**Coauthored Research Paper, submitted to Industrial Marketing Management (IMM)**, College Freshmen Year – Sophomore Year

- Coauthoring a research paper focusing on finding *the impact of supplier involvement on New Product Development (NPD) and product modularity in relation to governance mechanisms*
  - Performed statistical analysis on supplier involvement in product development to propose potential model and improvements for product modularity in supply chain modifications (linearity, EFA, CFA, etc.)
  - Used IBM Amos, SPSS, Excel Stats packet to make analysis and form models

**Internship at Matrix Origin Inc.** (July 14, 2025 – August 20, 2025)

- Interned at Matrix Origin Inc. as a Testing Intern under the Testing Department for 6 weeks
  - Contributed to the development and design of GenAI product and website through constructing test cases, proposing potential flaws and improvements, and testing product and website performance.

**Coauthored Research Paper, published by American Journal of Information Technology (AJIT),** Grade 11

- Coauthored conceptual paper titled *A Conceptual Framework of Matching Product Architecture and Customer Involvement in New Product Development (NPD): Role of Internet of Things (IoT)*.
  - Project sought to address the literature gap in the relationship between product architecture and customer involvement in New Product Development
  - Developed a conceptual framework for investigating the relationship between the timing/extent of customer involvement in NPD and the role of product architecture and Internet of Things (IoT)

**Independent Research Paper, published by Journal of High School Science,** Grade 11,12

- Completed research project titled *Predicting Affordability of a Family Paying Medical and Health Care Expenses – Development and Validation of a Predictive Model* and qualified for regional SEFH competition.
  - The project aimed to develop an applicable and reliable predictive model to predict whether a family can afford medical care based on 2021 National Survey of Children's Health (NSCH) dataset, which can be applied by insurance companies and welfare institutions to determine appropriate insurance premiums and levels of financial assistance to different families, thus helping to improve medical expense affordability.
  - Employed methods such as missing value imputation, dependent variable dichotomization, feature standardization, one-hot encoding, correlational analysis, multicollinearity testing, logistic regression, oversampling, regularization, and cross-validation to identify and model predictive factors with significant impacts on healthcare affordability.
  - Used model validation metrics – Confusion matrix, ROC curve, AUROC score, and Wald testing.

**HiMCM (High School Mathematical Contest in Modeling),** Grade 11

- Entered the competition with a group research paper that analyzed CO2 concentration levels over time, global warming resulting in temperature changes, and developed models to predict future CO2 levels and temperatures and won "International Honorable Mention".
  - Performed paired sample t-test on the increase of CO2 level over the given four decades to determine if there is a larger increase of CO2 concentration in last decade than in previous decades.
  - Developed three models to predict CO2 concentration levels, temperatures, and account for the relationship between CO2 and temperatures; and tested the accuracy of each model selected.

**Illustrator, Happy Earth Magazine (digital environmental magazine),** Grade 11, 12

- The purpose of Happy Earth Magazine is to raise environmental awareness and convey the need for climate action in young generations through journalism, creative writing, and arts.
- As an illustrator, I created cover art and digital illustrations for journal articles every month.

## Leadership & Volunteer Services

**Volunteer, CSSA Development Department,** College Freshmen Year
- CSSA is a non-profit org. that helps overseas Chinese students enrich their social, academic, and professional experiences, serving as a bridge between Chinese and local communities, and promoting cultural exchanges.
- By joining CSSA Development Department, I applied my skills to real-world projects, focusing on building technical solutions for CSSA, including website development, App design, back-end system construction, etc.

**Vice President, Art to Hearts Association (ATHA),** Grade 10, 11, 12
- ATHA is a teen association that supports children with special needs, sponsored by the non-profit organization Light & Salt Association, a caring center for people with special needs.
- As Vice President, recruited, trained, and led 30+ volunteers to teach 15+ special needs kids for specially designed art program. Organized and delivered training on how to connect with kids with special needs such as autism.
- As a volunteer, on a weekly basis, prepared and delivered weekly supplies needed for the program, produced instructional videos of art projects taught and assisted the special needs children in creating artworks and crafts.

**Vice President/Dance Leader, K-pop Club, Clements High School,** Grade 10, 11, 12
- K-pop Club is a student lead club that focuses on promoting Korean pop music, dance, and team spirit, and fostering a collaborative, positive atmosphere among students.
- As Vice President, recruited 70+ members, organized school concerts, led team performances at 13+ school/community festivals, won 1st place in community dance competition, and cultivated team spirit.
- As Dance Director, trained new dancers, assigned dance roles, planned, organized, and directed team practices.
- Performed at the Houston Korean Festival, K-pop Houston World Festival, school fundraising concerts, school World Culture Exchange Day, etc., and participated in various local dance competitions.

**Team Leader, Wharton Global High School Investment Competition,** Grade 10
- Organized and led a team of 4 from Clements High School to develop investment strategy and a suitable investment portfolio for a designated client.
- Worked with a professional equity analyst and a professional financial advisor on behalf of the team to better understand and analyze stock market flows and client needs analysis.
- Managed portfolio performance and achieved top 5% of l 5000+ teams by portfolio return.
- Completed a group investment strategy research report as the final deliverable of the competition.

## Languages & Tools
- Languages: English, Mandarin Chinese, French (beginner)
- Tools: Java, SPSS, Python, IBM Amos, Excel stats package, MediBang & Procreate (digital art), Microsoft Clipchamp (video editing)